

Wilson Harper

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wilsonharper.net

Houston, TX

Education

Rice University

Class of 2029

- B.S. Electrical Engineering; Minor in Entrepreneurship
- Lillian Haynie Trustee Scholar; Martel College Outstanding New Student Award

Experience

Electrical Engineering Intern, RTX BBN Technologies

Summer 2026

- Informed system architecture and analysis for a low-cost mobile embedded system within an agile team
- Evaluated radar range and detection ability on small mobile platforms through rigorous field testing
- Executed trade studies to select COTS boards and positioning systems under tight SWaP-C constraints
- Developed rugged field deployment solutions for high-speed, FPGA-based fiber-optic hardware systems
- Integrated in-house PCBs and COTS power supplies into operator-centric custom rackmount enclosures
- Designed a 4-layer PCB in Altium for harsh environments, integrating high-speed ICs and a robust PDN

Mission Control Lead, Rice Eclipse Rocketry

2025 - Present

- Leading development of electronics and software for Rice's liquid engine development program
- Executed the program's first successful liquid hotfire, enabling the transition to future liquid launches
- Integrating Raspberry Pi with custom PCBs, LabJack DAQ, and solenoids for critical control and data gathering
- Designed avionics architecture for an Active Flight Stabilization system intended for a spaceshot rocket
- Integrated RP2350 with I2C/UART/SPI sensors on a custom PCB to actuate servo canards in real-time

Undergraduate Researcher, Rice Experimental Pixels Lab

Spring 2026

- Engineered a compact, highly directional wind system to synchronize physical airflow with XR environments
- Designed and integrated remote blowers, ducting, and control circuitry within strict spatial constraints
- Implemented UDP with networked POE-powered ESP32s, servos, and PWM fans

Teaching Assistant, BUSI 361: Communication for Entrepreneurs

Spring 2026

- Facilitated instruction on communication, conflict resolution, networking, and workplace dynamics

Technical Projects

wlsn.ee/projects

Custom Rocket Flight Computer & Telemetry Stack

2026

- Designed, programmed, and integrated a custom RP2040-based flight computer in two weeks
- Engineered 1 Hz LoRa telemetry while logging altimeter, accelerometer, and GPS data at 20 Hz
- Built an accompanying ground station with custom antenna to display live data and Google Earth position

UAV with Position-Holding in GPS-Denied Environments

2022 - 2025

- Led national-competition team in design of UAV with optical flow and LiDAR to navigate in tight environments
- Implemented EKF sensor fusion, remote telemetry, modular airframe, and servo-actuated gripper

Doppler & Synthetic Aperture Radar at MIT Lincoln Laboratory RISE

Summer 2024

- One of ~4% of applicants selected for DoD-sponsored free research experience
- Researched, built, and programmed Doppler and synthetic aperture radars, mentored by Lincoln Labs staff

Skills

- **Certifications:** General Class Amateur Radio License, Tripoli Level 1 High Powered Rocketry Certification
- **Hardware & Systems:** LiDAR, radar, sensor integration, UAVs, PCB design, microcontrollers, LoRa, telemetry
- **Software:** Python, CAD (Fusion, Onshape), ArduPilot, Cloudflare, Tailscale, KiCad, Altium
- **Tools:** Oscilloscopes, soldering, 3D printing, hand tools, multimeters, breadboards, function generators